

5G/6G Communication Systems- ECA5601 (LAB)

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Experiment 8: Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim: To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

Objectives

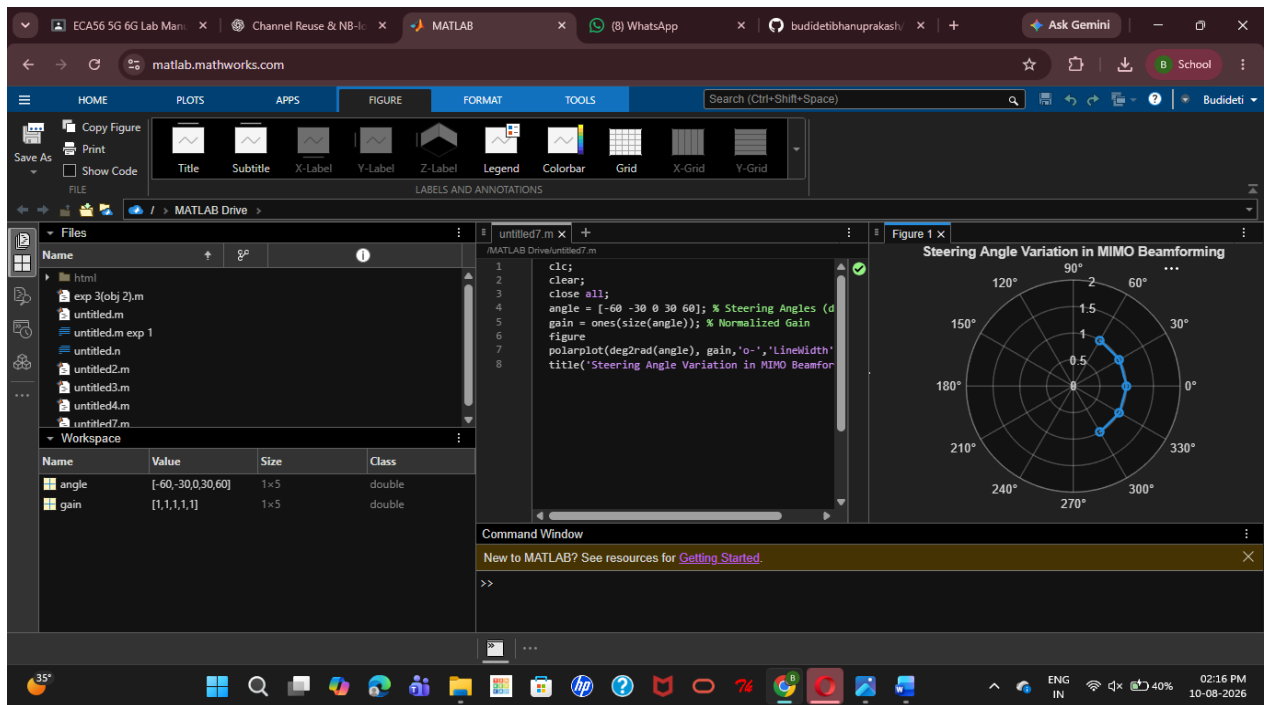
1. To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.
2. To compare the array gain achieved for different steering angles in a MIMO beamforming system using MATLAB.
3. To evaluate the beam direction error between the desired steering angle and the actual beam direction using MATLAB.

Objective 1:

Program:

```
clc; clear;close all;  
  
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)  
  
gain = ones(size(angle)); % Normalized Gain  
  
figure  
  
polarplot(deg2rad(angle), gain,'o-','LineWidth',2)  
  
title('Steering Angle Variation in MIMO Beamforming')
```

Output:



Objective 2:

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
```

```
gain = [9 12 15 12 9]; % Array Gain (dB)
```

```
figure
```

```
bar(angle, gain)
```

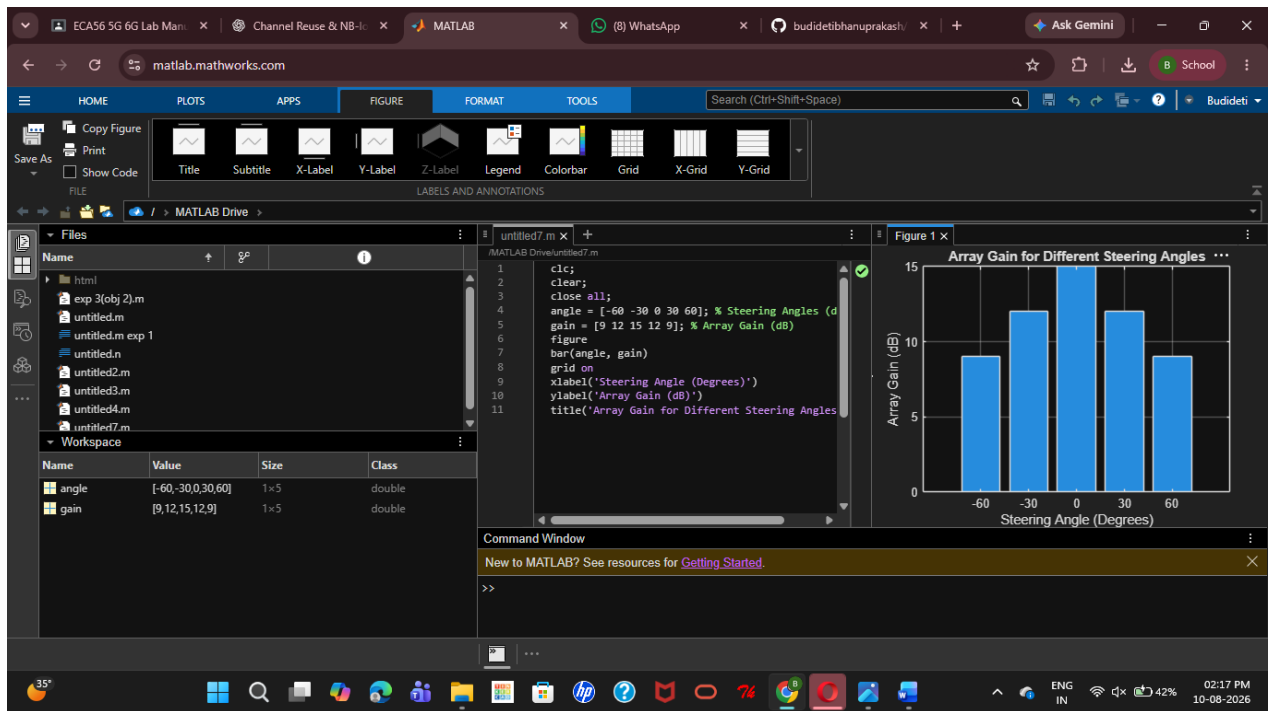
```
grid on
```

```
xlabel('Steering Angle (Degrees)')
```

```
ylabel('Array Gain (dB)')
```

```
title('Array Gain for Different Steering Angles')
```

Output:



Objective 3:

MATLAB Code:

```

clc;

clear;

close all;

desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees)
actual = [-58 -32 1 29 62]; % Actual Beam Direction (degrees)
error = abs(desired - actual); % Beam Direction Error

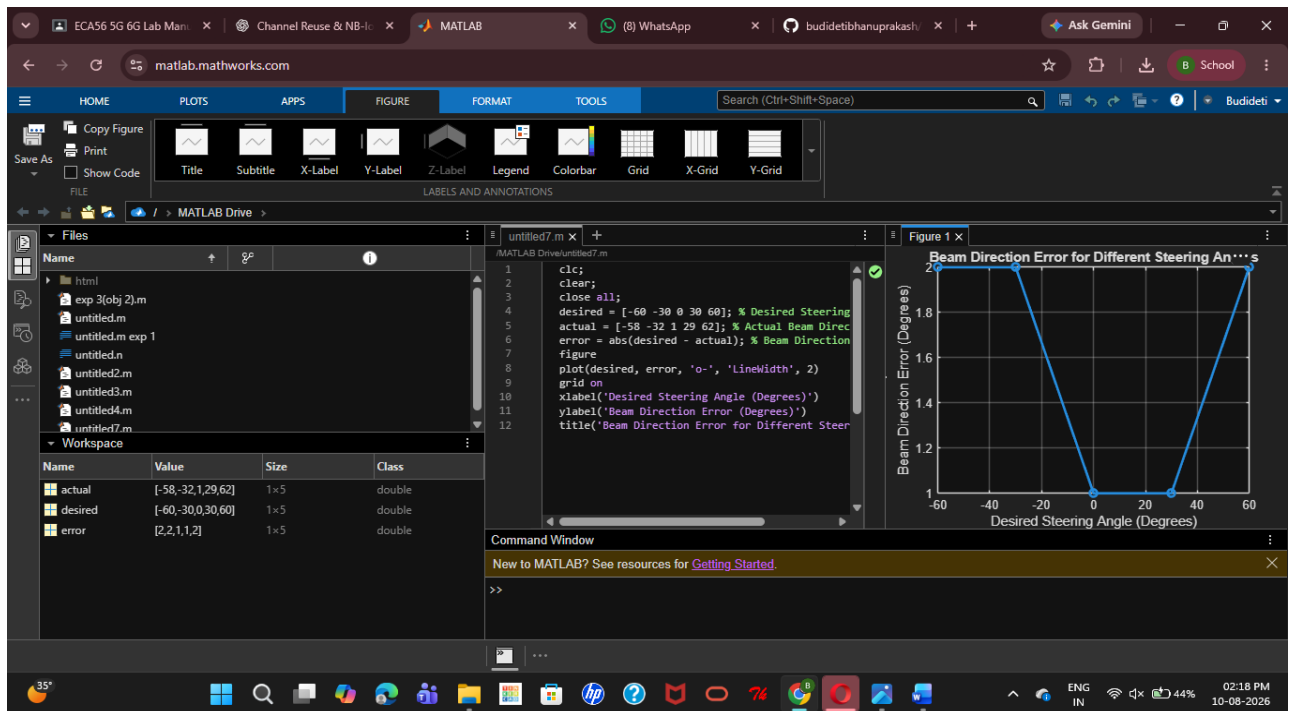
figure
plot(desired, error, 'o-', 'LineWidth', 2)

grid on

xlabel('Desired Steering Angle (Degrees)')
ylabel('Beam Direction Error (Degrees)')
title('Beam Direction Error for Different Steering Angles')

```

Output:



Result:

The MIMO beamforming system was successfully analysed, showing that steering angle controls beam direction, array gain varies with steering angle, and beam direction error can be evaluated accurately using MATLAB.